A product counterexample to the simplex maximum for projection-body volume

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Abstract

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum.

1 The result

Conventions. Throughout, $d \geq 2$. A *convex body* in $\mathbb{R}^d$ is a compact convex set with nonempty interior, and $|K|$ is its d-dimensional volume. For compact convex C, $h_C(x) = \max_{y \in C} \langle x, y \rangle$ is its *support function*. We write $C + D = \{c + d : c \in C,\ d \in D\}$ for the Minkowski sum, S^{d-1} for the unit sphere, B_2^j for the unit ball of $\mathbb{R}^j$, $\kappa_j = |B_2^j|$, and $\mathrm{proj}_{u^\perp}$ for orthogonal projection onto $u^\perp$. Three standard facts are used.

(S1) $h_{C+D} = h_C + h_D$; $h_{C \times D}(u, v) = h_C(u) + h_D(v)$; $h_{MC}(x) = h_C(M^t x)$ for linear M.

(S2) h_C determines C, and $C \subset D$ iff $h_C \leq h_D$. (If $x \notin C$ and $y \in C$ is nearest to x, then $\langle x - y, z - y \rangle \leq 0$ for all $z \in C$, so $\langle x - y, \cdot \rangle$ separates x from C.)

(S3) $h_{[-v/2, v/2]}(u) = \frac{1}{2} |\langle v, u \rangle|$.

Renamed symbols: the source's R (a radius), $F(u)$, λ_m, A (a constant) and d_r are here ρ, $g(u)$, μ_m, α and q, to avoid clashes.

Definitions. The *projection body* ΠK (also $\Pi_d K$) of a convex body $K \subset \mathbb{R}^d$ is the compact convex set with

$$h_{\Pi K}(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} K), \qquad u \in S^{d-1},$$

extended to $\mathbb{R}^d$ positively homogeneously of degree one. (Existence: Lemma 2 for polytopes; Lemma 4, Step 4, in general.) We study

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \qquad c_d = \frac{(d+1) d^d}{d!}. \tag{1}$$

Brannen conjectured that simplices maximize R_d among all convex bodies [1]. Every simplex has $R_d = c_d$ (Proposition 5), so the conjecture reads

$$|\Pi K| \leq c_d |K|^{d-1}. \tag{2}$$

Theorem 1. *Let $T_{10} = \mathrm{conv}(0, e_1, \ldots, e_{10}) \subset \mathbb{R}^{10}$, where the e_i are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Thus K violates (2) in dimension twenty.

Corollary 6 shows that products of simplices exceed c_n by a factor λ^n, $\lambda > 1$, in all large dimensions n.

Context. Petty established the affine covariance of Π [6]; see also [5, Equation (1)]. The functional involves $|\Pi K|$, not the volume of the polar; for $d \geq 3$, Petty's minimum problem asks whether ellipsoids are the only minimizers of R_d [7, Section 1]. In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and prove $R_3(K) \leq 18 = c_3$, with tetrahedra attaining equality [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d+2$ facets [3, Theorem 1.3], via perturbed projections of a cube and Minkowski's existence theorem [3, Section 4]. So the conjecture was already refuted; the present contribution is a direct product construction with exact value. For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]. The optimal upper constant and its maximizers remain separate questions.

2 Proof at a glance

Key idea. R is multiplicative over Cartesian products of polytopes, but the simplex constants are not: $c_{10}^2 > c_{20}$.

1. Lemma 2 (facet formula): for a polytope, ΠP is a sum of segments, one per facet. Everything below rests on it.
2. Proposition 3: the facets of $A \times B$ are $F \times B$ and $A \times G$; Lemma 2 then gives $\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B)$ and $R_{r+s}(A \times B) = R_r(A)R_s(B)$.
3. Proposition 5: by Lemma 2, ΠT_d is a scaled cube plus one segment, of volume $(d+1)/((d-1)!)^d$; hence $R_d(T_d) = c_d$.
4. Theorem 1 follows from Propositions 3 and 5 by integer arithmetic.
5. Lemma 4 (affine invariance of R_d) is used only for the last sentence of Proposition 5: *every* simplex has value c_d. The displayed identity of Theorem 1 does not depend on it.
6. Corollary 6 iterates Proposition 3 and uses $c_n < (n+1)(11/4)^n$ with $c_{10} > (11/4)^{10}$.

3 The facet formula

This is the polytopal case of Cauchy's projection formula [7, Section 1]; the proof is included to fix the factor $1/2$.

Lemma 2. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. For each facet F, let s_F be its $(d-1)$-dimensional volume and ν_F its outward unit normal. Then*

$$h_{\Pi P}(u) = \frac{1}{2} \sum_F s_F |\langle \nu_F, u \rangle|, \qquad u \in \mathbb{R}^d, \tag{3}$$

and therefore

$$\Pi P = \sum_F \left[-\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right]. \tag{4}$$

Proof. Plan. Each facet's shadow has volume $s_F|\langle \nu_F, u \rangle|$, and the shadows cover the shadow of P exactly twice almost everywhere. Fix a unit vector u and write $\pi = \mathrm{proj}_{u^\perp}$.

Step 1: $\mathrm{vol}_{d-1}(\pi F) = s_F|\langle \nu_F, u\rangle|$. Let $v_1, \dots, v_{d-1}$ be an orthonormal basis of $\nu_F^\perp$. On the hyperplane of F, π is affine with Jacobian the $(d-1)$-volume of the parallelepiped spanned by $\pi v_i = v_i - \langle v_i, u\rangle u$ in $u^\perp$. This equals $|\det(\pi v_1, \dots, \pi v_{d-1}, u)| = |\det(v_1, \dots, v_{d-1}, u)|$ by column operations. Substituting $u = \langle u, \nu_F\rangle \nu_F + \sum_i \langle u, v_i\rangle v_i$ gives $|\langle u, \nu_F\rangle|\,|\det(v_1, \dots, v_{d-1}, \nu_F)| = |\langle \nu_F, u\rangle|$.

Step 2: almost every $y \in \pi P$ lies in πF for exactly two facets F. Let $Z \subset u^\perp$ be the union of the relative boundary of πP, the images of all faces of dimension at most $d-2$, and the images of facets with $\langle \nu_F, u\rangle = 0$. Then Z is null: the first set is the boundary of a convex set, and the others lie in finitely many affine subspaces of dimension at most $d-2$. Take $y \in \pi P \setminus Z$ and $\ell = y + \mathbb{R}u$. Since $\pi(\operatorname{int} P)$ is an open convex set with the same closure as πP, it equals the interior of πP. So ℓ meets $\operatorname{int} P$, and $\ell \cap P = [p, q]$ with $p \neq q$ and $\ell \cap \partial P = \{p, q\}$. Neither endpoint lies in a face of dimension at most $d-2$, so each lies in exactly one facet; call them F_p, F_q. If F is any facet with $y \in \pi F$, then F is not parallel to u, so its hyperplane meets ℓ in a single point, which lies in $F \subset \partial P$ and is therefore p or q. Hence $F \in \{F_p, F_q\}$, and $F_p \neq F_q$ because one such hyperplane cannot contain both p and q.

Step 3. Integrating $\sum_F \mathbf{1}_{\pi F} = 2 \cdot \mathbf{1}_{\pi P}$ (valid off Z) over $u^\perp$ and using Step 1 gives $\sum_F s_F|\langle \nu_F, u\rangle| = 2\,\mathrm{vol}_{d-1}(\pi P)$, which is (3) for unit u. Both sides are positively homogeneous of degree one, so (3) holds on $\mathbb{R}^d$.

Step 4. By (S3) and (S1), the right side of (3) is the support function of the right side of (4); (S2) gives (4), and in particular ΠP exists. □

4 Cartesian products

Proposition 3. *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \tag{5}$$

In particular,

$$R_{r+s}(A \times B) = R_r(A) R_s(B). \tag{6}$$

This is the engine of Theorem 1 and Corollary 6.

Proof. Step 1: the facets of $A \times B$ are $F \times B$ and $A \times G$, for facets F of A and G of B. The functional (u, v) attains its maximum over $A \times B$ at (a, b) exactly when a maximizes $\langle u, \cdot\rangle$ on A and b maximizes $\langle v, \cdot\rangle$ on B. So the face exposed by (u, v) is the product $A_u \times B_v$ of the exposed faces, with $A_0 = A$, $B_0 = B$. Every facet is exposed by its normal. If $u \neq 0 \neq v$, both factors are proper faces and $\dim(A_u \times B_v) \leq (r-1) + (s-1)$, so this is not a facet. If $v = 0$, then $\dim(A_u \times B) = \dim A_u + s$, which is $r + s - 1$ iff A_u is a facet of A; symmetrically for $u = 0$.

Step 2. The facet $F \times B$ has outward unit normal $(\nu_F, 0)$ and, the two coordinate spaces being orthogonal, $(r+s-1)$-volume $s_F|B|$. Likewise $A \times G$ has normal $(0, \nu_G)$ and volume $|A|s_G$.

Step 3. Lemma 2, applied to $A \times B$, then to A and B, gives

$$h_{\Pi(A\times B)}(u, v) = \tfrac{1}{2}\sum_F |B| s_F |\langle \nu_F, u\rangle| + \tfrac{1}{2}\sum_G |A| s_G |\langle \nu_G, v\rangle| = |B| h_{\Pi A}(u) + |A| h_{\Pi B}(v).$$

By (S1) this is the support function of the right side of (5); (S2) gives (5).

Step 4. Since $\Pi A \subset \mathbb{R}^r$ and $\Pi B \subset \mathbb{R}^s$, (5) gives $|\Pi(A \times B)| = |B|^r |\Pi A| \cdot |A|^s |\Pi B|$. With $|A \times B| = |A||B|$,

$$R_{r+s}(A \times B) = \frac{|B|^r |A|^s |\Pi A||\Pi B|}{(|A||B|)^{r+s-1}} = \frac{|\Pi A|}{|A|^{r-1}} \cdot \frac{|\Pi B|}{|B|^{s-1}}.$$ □

5 Affine invariance

Lemma 4. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. For every invertible linear map $L\colon \mathbb{R}^d \to \mathbb{R}^d$,*

$$\Pi(LP) = |\det L| L^{-t} \Pi P, \qquad R_d(LP) = R_d(P),$$

where L^{-t} denotes the inverse transpose. Translations also preserve R_d.

Used only in the last sentence of Proposition 5, and there only for polytopes (Steps 1–3).

Proof. Put $M = |\det L| L^{-t}$. First let P be a polytope; the facets of LP are the sets LF.

Step 1: the area-normal vector of LF is $M(s_F \nu_F)$. Let $n_F = L^{-t}\nu_F / \|L^{-t}\nu_F\|$. For $x \in P$, $y \in F$ we have $\langle Lx - Ly, L^{-t}\nu_F \rangle = \langle x - y, \nu_F \rangle \leq 0$, with equality exactly on the hyperplane of F; so n_F is the outward unit normal of LF. The prism $F + [0, \nu_F]$ has volume s_F. Its image $LF + [0, L\nu_F]$ has volume $|\det L| s_F$ and height $\langle L\nu_F, n_F \rangle = 1/\|L^{-t}\nu_F\|$ over LF. Dividing, $s_{LF} = |\det L| \, \|L^{-t}\nu_F\| \, s_F$, so $s_{LF} n_F = M(s_F \nu_F)$, whatever the sign of $\det L$.

Step 2. By (4) for LP and for P, $\Pi(LP) = \sum_F M[-s_F\nu_F/2,\, s_F\nu_F/2] = M\,\Pi P$.

Step 3. $|\det M| = |\det L|^d |\det L|^{-1} = |\det L|^{d-1}$, so $|\Pi(LP)| = |\det L|^{d-1} |\Pi P|$, while $|LP|^{d-1} = |\det L|^{d-1} |P|^{d-1}$. Hence $R_d(LP) = R_d(P)$. Projections of $P + t$ are translates of those of P, so $\Pi(P + t) = \Pi P$ and $|P + t| = |P|$.

Step 4: arbitrary convex bodies (not needed for Theorem 1 or Corollary 6). Translate so that $0 \in \operatorname{int} P$ and fix $\rho > 0$ with $P \subset \rho B_2^d$. Choose polytopes $P_1 \subset P_2 \subset \cdots \subset P$, all containing a fixed ball aB_2^d, $a > 0$, with $P \subset P_m + \varepsilon_m B_2^d$ and $\varepsilon_m \to 0$. For example, take convex hulls of a fixed inscribed polytope containing aB_2^d together with cumulative finite ε_m-nets of P. Since $\varepsilon_m B_2^d \subset (\varepsilon_m/a) P_m$ and P_m is convex,

$$P_m \subset P \subset \mu_m P_m, \qquad \mu_m = 1 + \varepsilon_m / a \longrightarrow 1. \tag{7}$$

For unit u put $g(u) = \operatorname{vol}_{d-1}(\operatorname{proj}_{u^\perp} P)$ and $f_m(u) = h_{\Pi P_m}(u)$. Projecting (7) gives

$$f_m \leq g \leq \mu_m^{d-1} f_m, \tag{8}$$

and $f_m \leq \kappa_{d-1} \rho^{d-1}$ because $P_m \subset \rho B_2^d$. Hence $0 \leq g - f_m \leq (\mu_m^{d-1} - 1)\kappa_{d-1}\rho^{d-1}$, so $f_m \to g$ uniformly. The bodies ΠP_m increase by (S2), lie in a fixed ball, and contain $\kappa_{d-1} a^{d-1} B_2^d$ because $P_m \supset aB_2^d$. The closure of their union is therefore a convex body with support function $\sup_m f_m = g$; this is ΠP. By (8) and (S2),

$$\Pi P_m \subset \Pi P \subset \mu_m^{d-1} \Pi P_m, \qquad |\Pi P_m| \leq |\Pi P| \leq \mu_m^{d(d-1)} |\Pi P_m|,$$

and (7) gives $|P_m| \leq |P| \leq \mu_m^d |P_m|$. Applying L to (7) gives the same statements for LP_m and LP. By Step 2 and (S1), $h_{\Pi(LP_m)}(x) = |\det L| \, h_{\Pi P_m}(L^{-1}x)$; letting $m \to \infty$ gives the same identity for P, which is the covariance. The volume inequalities pass $R_d(LP_m) = R_d(P_m)$ to the limit. □

6 The simplex value

Proposition 5. *For $d \geq 2$, the standard simplex $T_d = \operatorname{conv}(0, e_1, \ldots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \qquad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \qquad R_d(T_d) = c_d.$$

Every full-dimensional d-simplex has the same value c_d.

Proof. Plan. List the $d+1$ area-normal vectors, recognise ΠT_d as a scaled cube plus one segment, and compute that volume by slicing. Put $w = e_1 + \cdots + e_d$ and $Q = [0,1]^d$.

Step 1. T_d is the cone over $T_{d-1} \times \{0\}$ with apex e_d at height one, so $|T_d| = |T_{d-1}|/d$; with $|T_1| = 1$ this gives $|T_d| = 1/d!$.

Step 2: area-normal vectors. $T_d = \{x : x_i \geq 0,\ \sum x_i \leq 1\}$. The facet in $\{x_i = 0\}$ is a copy of T_{d-1} with outward normal $-e_i$, so its area-normal vector is $-e_i/(d-1)!$. The remaining facet $F_0 = \operatorname{conv}(e_1, \ldots, e_d)$ has outward unit normal $w/\sqrt{d}$. Dropping the last coordinate maps F_0 bijectively onto T_{d-1} and, by Step 1 of Lemma 2 with $u = e_d$, multiplies area by $|\langle w/\sqrt{d}, e_d\rangle| = 1/\sqrt{d}$. So $s_{F_0} = \sqrt{d}/(d-1)!$ and the area-normal vector is $w/(d-1)!$. (Check: the $d+1$ vectors sum to zero.)

Step 3. A centered segment is unchanged when its generator changes sign, so (4) gives

$$\Pi T_d = \frac{1}{(d-1)!}\Big(\sum_{i=1}^{d}\big[-\tfrac{e_i}{2}, \tfrac{e_i}{2}\big] + \big[-\tfrac{w}{2}, \tfrac{w}{2}\big]\Big),$$

which is a translate (by $-w/(d-1)!$) of $(Q + [0,w])/(d-1)!$, since $Q = \sum_i [0, e_i]$.

Step 4: $|Q + [0,w]| = d+1$. Let $\theta = w/\|w\|$. For $y \in w^\perp$, the fibre $\{t : y + t\theta \in Q\}$ is empty or an interval $[a,b]$. The corresponding fibre of $Q + [0,w]$ is then empty or $[a, b + \|w\|]$, so both sets have the same projection to $w^\perp$. By Fubini and homogeneity of $h_{\Pi Q}$,

$$|Q + [0,w]| = |Q| + \|w\| \operatorname{vol}_{d-1}(\operatorname{proj}_{w^\perp} Q) = 1 + h_{\Pi Q}(w).$$

The cube has facets of area one with normals $\pm e_i$, so (3) gives $h_{\Pi Q}(w) = \frac{1}{2}\sum_i 2|\langle e_i, w\rangle| = d$. (Check, $d = 2$: $Q + [0,(1,1)]$ is the hexagon with vertices $(0,0), (1,0), (2,1), (2,2), (1,2), (0,1)$, of area 3.)

Step 5. Steps 3 and 4 give $|\Pi T_d| = (d+1)/((d-1)!)^d$. With Step 1 and $d! = d\,(d-1)!$,

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)d^{d-1}}{(d-1)!} = \frac{(d+1)d^d}{d!} = c_d.$$

(Check: $c_2 = 6$, and $c_3 = 18$ matches the three-dimensional bound quoted in Section 1.)

Step 6. Every full-dimensional d-simplex is $LT_d + t$ with L invertible, so Lemma 4 gives it the value c_d. □

7 The counterexample and exponential excess

Proof of Theorem 1. K is a full-dimensional convex polytope in $\mathbb{R}^{20}$. Proposition 3 with $A = B = T_{10}$, then Proposition 5, give

$$R_{20}(K) = R_{10}(T_{10})^2 = c_{10}^2 = \left(\frac{11 \cdot 10^{10}}{10!}\right)^2.$$

Since $c_{20} = 21 \cdot 20^{20}/20!$, $20^{20} = 2^{20}10^{20}$ and $20!/(10!)^2 = \binom{20}{10}$,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}}.$$

Finally $\binom{20}{10} = 184\,756$ and $2^{20} = 1\,048\,576$, so the numerator is $22\,355\,476$, the denominator is $22\,020\,096$, and

$$121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0.$$

□

General product test. For $r, s \geq 2$, the same two propositions and $c_r c_s / c_{r+s}$ written out give

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1} \binom{r+s}{r} \frac{r^r s^s}{(r+s)^{r+s}}.$$

Thus any universal upper constant in dimension $r+s$ must be at least $c_r c_s$.

Corollary 6. *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

The polytopes K_n may be chosen as Cartesian products of simplices.

Proof. Plan. Repeated ten-dimensional factors supply exponential growth; one extra simplex of dimension 10 to 19 reaches every $n \geq 20$; a crude bound $c_n < (n+1)\beta^n$ controls the denominator.

Step 1: construction. Every integer $n \geq 20$ is uniquely $n = 10k + q$ with $q \in \{10, \ldots, 19\}$ and $k \geq 1$. Let $K_n = T_{10}^k \times T_q$, where T_{10}^k is the k-fold Cartesian product. Each partial product is a full-dimensional polytope of dimension at least ten, so Proposition 3 applies k times; with Proposition 5,

$$R_n(K_n) = c_{10}^k \, c_q, \qquad R_n(T_n) = c_n.$$

Step 2: $e < \beta := 11/4$. In $e = \frac{5}{2} + \sum_{j \geq 3} \frac{1}{j!}$, the tail starts at $\frac{1}{6}$ and has successive ratios $\frac{1}{j+1} \leq \frac{1}{4} < \frac{1}{3}$. So the tail is less than $\frac{1}{6} \sum_{j \geq 0} 3^{-j} = \frac{1}{4}$.

Step 3: $c_n < (n+1)\beta^n$ for $n \geq 2$. The number $n^n/n!$ is one term of the positive series $e^n = \sum_j n^j/j!$. Hence $c_n = (n+1)n^n/n! < (n+1)e^n < (n+1)\beta^n$.

Step 4: $\alpha := c_{10}/\beta^{10} > 1$. Since $10! = 6400 \cdot 567$, $c_{10} = 11 \cdot 10^{10}/10! = 17\,187\,500/567$. Then, cancelling one factor 11,

$$\alpha = \frac{17\,187\,500 \cdot 4^{10}}{567 \cdot 11^{10}} = \frac{1\,562\,500 \cdot 2^{20}}{567 \cdot 11^9} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1.$$

Step 5. Fix λ with $1 < \lambda < \alpha^{1/10}$. By Steps 1 and 3, and $n = 10k + q$,

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} = \frac{c_{10}^k c_q}{c_n \lambda^n} > \frac{c_{10}^k c_q}{(n+1)\beta^n \lambda^n} = \frac{c_q}{\beta^q \lambda^q} \cdot \frac{(\alpha/\lambda^{10})^k}{10k+q+1}.$$

For fixed q the right side tends to infinity as $k \to \infty$, because $\alpha/\lambda^{10} > 1$. So there is k_q with right side at least 1 for all $k \geq k_q$. Let $N = \max\bigl(20, \max_q(10k_q + q)\bigr)$. If $n = 10k + q \geq N$ then $k \geq k_q$, so the ratio is at least λ^n. □

8 Verification checklist

1. **Factor $\frac{1}{2}$ and the double cover** (Lemma 2, Steps 1–3): the Jacobian $|\langle \nu_F, u \rangle|$, and that almost every line parallel to u through P meets exactly two facets.
2. **Facets of a product** (Proposition 3, Steps 1–2): only $F \times B$ and $A \times G$ occur, with areas $s_F|B|$ and $|A|s_G$.
3. **Exponent bookkeeping** (Proposition 3, Step 4): $|B|^r |A|^s$ against $(|A||B|)^{r+s-1}$.
4. **Simplex computation** (Proposition 5, Steps 2 and 4): the slanted facet has area-normal vector $w/(d-1)!$, and $|Q + [0, w]| = d + 1$.

5. **Arithmetic of Theorem 1**: the reduction to $121\binom{20}{10}/(21\cdot 2^{20})$ and $121\cdot 184\,756 > 21\cdot 1\,048\,576$.
6. **Corollary 6**: $e < 11/4$ (Step 2), the value of α (Step 4), and the exponent split $n = 10k + q$ in Step 5.
7. **Scope of Lemma 4**: Steps 1–3 (polytopes) suffice for "every simplex has value c_d"; Step 4 only extends the lemma to general bodies.

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